

Foundation Program in Literacy, Numeracy and Skills

Numeracy - Grade 8

Lesson 9 and Lesson 11: Teacher's Guide

Circles



A: Understanding how to measure and calculate the radius, diameter and circumference of circles and apply these skills to real life situations

B. Competencies

Application of geometry in practical situations and trades

Element 4: Find circumference and area of a circle

Performance Criteria: Level 1: Find the radius, diameter, and circumference of a circle

C. Learning Outcomes

At the end of the lesson, students should be able to identify and define radius, diameter and circumference of a circle

D. Resources and Materials:

- String or flexible measuring tape
- Circular objects (plates, cans, lids, bottle caps, round table)
- Rulers
- Whiteboard/Markers
- Calculators
- Compass box

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E. Implementation Guidelines

Activity 1:

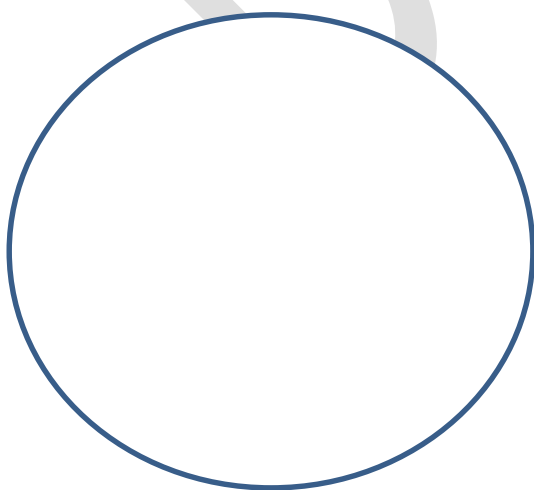
Introduction (Group discussion)

- Brainstorm about circles all around us
- Show students a variety of circular objects and give counter examples like sphere, oval are not circles.
- Discuss what they know about circles.
- Introduce the terms: Radius, diameter and circumference using real life situations. Draw diagrams to show the above visually
- Take a small plate. With a pencil, make an outline of the plate on a sheet of A4 paper. The trace that is made is a **circle**.
- Cut the circle out of the paper and fold it so that you get two equal parts
- Open the shape now. The line you get is called the **diameter** of the circle.
- Fold the circle into two equal parts along another line. This gives another diameter.

We observe the following:

- (a) The place where two diameters meet is called the **centre** of the circle.
- (b) The **diameter** is a straight line passing through the centre.
- (c) The **radius** is the distance from the centre to a point on the circle.
It is measured in **mm, cm** or **m**.
- (d) The diameter of a circle = 2 x radius

Discuss how we can locate the center of a circle



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Activity 2:

- Divide the class into small groups
- Take a can of soft drink. Place a strip of paper and stick around the can. Do not overlap the two ends of the strip of paper.
- Ask students to remove the strip and measure its length using a ruler as shown below
- When placed around a paper cup, the strip measures the circumference of the paper cup.

Circumference of paper cup = cm

Circumference of a circle is the distance all around it.

- Look for objects around you and record their radius, diameter and circumference.
- On a worksheet, make the students record their findings as follows:

Object	Radius	Diameter	Circumference
Plate			
Clock			
Bottle caps			
Cans			
Round Table			

- Make the students discuss their findings.

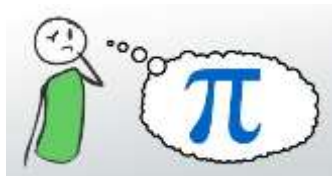
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Teacher's Guide: GRADE 8- Lesson 11

Circle

Activity 3: Introduction of π and formula to calculate circumference of circle



- Take three cans of different sizes.



- Place each one on a sheet of paper and trace a circle.
- Measure the diameter and record your answer in the first column marked (A) in the table below.
- Measure strips of paper to find the circumference of each can. Again, record your answer in the second column marked (B) in the table:

	Diameter (A)	Circumference(B)	<u><i>circumference</i></u> <u><i>diameter</i></u>
A			
B			
C			

Teacher makes the students use a calculator to find the value of

$$\frac{\text{circumference}}{\text{diameter}}$$

The value obtained in each case is close to **3**. This is an approximate value.

A common approximate value which is used in all books is $\frac{22}{7}$ or 3.142.

The exact value is not known but is given by the symbol π (Pi).

Teacher shows the students how to find the value of π on a calculator.



Circumference of circle = $2 \times \text{radius} \times \pi$

Circumference of circle = $\text{diameter} \times \pi$

- Teacher uses an example to show how to use formula for circumference of circle given radius (Students have to use calculator)

Example:

Find the circumference of a circle having radius = 5 cm

$$\begin{aligned}\text{Circumference of circle} &= 2 \times \text{radius} \times \pi \\ &= 2 \times 5 \times \pi \\ &= 31.4 \text{ cm}\end{aligned}$$

F. ASSESSMENT (in worksheet)

G. EXTENSION ACTIVITY

Area of circle